

WTW158 Worksheet on integration 1

Evaluate the following integrals WITHOUT using substitution.

1. $\int (\pi + \frac{1}{x}) dx$
2. $\int (z^4 + 4^z + e^{4z} + 4^{ez}) dz$
3. $\int x(x^2 + 1)^6 dx$
4. $\int \frac{6x^2}{(x^3 + 1)^6} dx$
5. $\int (4x + 6x^2) e^{x^2+x^3} dx$
6. $\int \frac{e^{2x}}{\sqrt{1 - e^{2x}}} dx$
7. $\int \frac{x e^{\arctan(x^2)}}{1 + x^4} dx$
8. $\int \sin(30 + 4x^3) x^2 dx$
9. $\int x^2 \operatorname{cosec}(x^3) \cot(x^3) dx$
10. $\int \sec(\sin 2x) \tan(\sin 2x) \cos 2x dx$
11. $\int \frac{\cos(\ln x)}{x} dx$
12. $\int \frac{x^9}{1 + x^{10}} dx$
13. $\int \frac{x^4}{1 + x^{10}} dx$
14. $\int \frac{1}{(1 + 4x^2) \arctan 2x} dx$
15. $\int \frac{x^2 \arcsin x^3}{\sqrt{1 - x^6}} dx$
16. $\int \frac{x^3}{\sqrt{1 - x^8}} dx$
17. $\int \frac{x^7}{\sqrt{1 - x^8}} dx$
18. $\int \frac{e^{-x}}{\sqrt{1 - e^{-2x}}} dx$
19. $\int 6^{2t} dt$
20. $\int 3^y \sqrt{1 - 3^y} dy$

DEFINITE INTEGRALS AND AREAS

Evaluate the following definite integrals.

1. $\int_{-2}^{-1} (x - 5) dx$

2. $\int_{-2}^{-1} \frac{1}{x} dx$

3. $\int_{-2}^{-1} \frac{1}{x^2} dx$

4. $\int_0^{\frac{\pi}{2}} (\sin x - 4 \cos 3x) dx$

5. $\int_{-2}^{-1} \frac{x}{9 + x^2} dx$

6. $\int_0^1 \frac{1}{9 + x^2} dx$

7.1 Sketch the graph of $f(x) = x^2 + x - 2$ on the interval $[-1, 2]$

7.2 Find the area of the region between the graph of f , the x -axis and the lines $x = -1$ and $x = 2$.

8. Consider the function $f(x) = xe^{x^2}$

8.1 Evaluate $\int f(x) dx$

8.2 Evaluate $\int_{-1}^1 f(x) dx$

8.3 Prove that $y = f(x)$ is an odd function.

8.4 Find, WITHOUT sketching the graph of the function, the area of the region between the graph of f , the x -axis and the lines $x = -1$ and $x = 1$

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